Review Article

# Comprehensive Review on Low-Level Laser Therapy in Dentistry

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ABSTRACT

Low-level laser therapy (LLLT) has broad applications in dentistry, enhancing patient outcomes through its anti-inflammatory, analgesic, and biostimulatory effects. It is used for reducing pain and swelling, accelerating wound healing, and promoting tissue regeneration. LLLT is effective in managing conditions, such as temporomandibular disorders (TMDs), oral mucositis, and postoperative discomfort. Additionally, it aids in orthodontics by accelerating tooth movement and mitigating associated pain. This non-invasive treatment offers a safe, efficient, and clinically proven approach, integrating well into various dental procedures to improve overall patient care and recovery.

**KEYWORDS:** *Dentistry, low-level laser therapy, LLLT*

## INTRODUCTION

Low-level laser therapy (LLLT) has emerged as a significant innovation in dentistry due to its therapeutic potential and non-invasive nature. Introduced by Endre Mester in Hungary during the early 1970s, LLLT initially faced skepticism, especially in Western countries. However, advancing research has revealed its biological and medical benefits, leading to widespread acceptance and integration into clinical practices worldwide.[1]

LLLT involves using low-level lasers to stimulate tissue healing and reduce pain and inflammation without causing significant heat or damage. Unlike high-powered lasers used for cutting or ablating tissues, LLLT operates at a lower energy level, targeting cellular components to promote regeneration and improve cellular function. Dentistry, in particular, has widely adopted LLLT, enhancing various procedures to improve patient care. It alleviates pain and inflammation in conditions, such as temporomandibular joint disorders, accelerates tissue repair post-surgery, and manages oral lesions. LLLT's therapeutic effects—such as pain relief, reduced inflammation, and faster healing—have transformed

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patient experiences and outcomes in dental treatments. As scientific evidence supporting LLLT's efficacy continues to grow, it is increasingly featured in major clinical laser conferences and published in reputable American and European journals. Often referred to as "mid-laser" or "soft-laser," this therapy is characterized by its gentle yet effective impact on oral tissues, performing without inducing macroscopic changes in tissue structure.[1,2]

## Mechanism of LLLT

Laser irradiation's biostimulatory effects induce physiological, metabolic, and functional changes in living microorganisms, with the 1-mW HeNe laser being the first commercially available biostimulation laser. In dentistry, the gallium-aluminum-arsenide laser, operating at wavelengths of 780 and 830 nm, is the most utilized for LLLT. Photon absorption by cellular photoacceptors triggers shifts in molecular configurations, leading to changes in cellular signaling. These primary reactions are followed by secondary reactions that modify cellular functions and signaling pathways, with reports indicating that laser-enhanced biostimulation accelerates intracellular metabolic processes, such as matrix production, fibroblast migration, and cell division. The laser's effectiveness in stimulating biological processes depends on the amount of light energy absorbed, which is influenced by factors, such as wavelength, optical and thermal properties, intensity, energy output, exposure duration, and the tissue's absorption and scattering coefficients.[1,3]

## Biological impact of LLLT

LLLT stimulates cellular activity and enhances tissue repair by increasing ATP production and cellular metabolism. It reduces inflammation, edema, and pain through photochemical reactions in cells. Benefits include accelerated wound healing, increased collagen production, and improved blood circulation. LLLT's effectiveness is influenced by factors, such as wavelength, dose, and tissue type. Commonly used in physical therapy, dentistry, and dermatology, LLLT helps treat various conditions, such as musculoskeletal pain, soft tissue injuries, and chronic joint disorders. It exerts minimal side effects, making it a safe, non-invasive therapeutic option for promoting tissue regeneration and reducing inflammation.[1,3,4]

## LLLT in oral surgical procedures

LLLT is highly beneficial in oral surgeries, reducing postoperative pain and discomfort, thus promoting faster recovery. It minimizes inflammation and swelling, aiding quicker healing. By stimulating cellular activity and angiogenesis, it enhances tissue repair and wound closure. Additionally, its antimicrobial properties help reduce infection risks. Consequently, LLLT is increasingly adopted in oral surgery to improve patient outcomes and recovery.[1]

## LLLT in endodontics

In endodontics, LLLT offers significant advantages. It reduces pain and inflammation associated with root canal treatments, promoting quicker patient relief. LLLT stimulates the healing of periapical tissues and enhances bone regeneration, improving overall treatment success. Additionally, its antimicrobial properties help eliminate bacteria in root canals, reducing the risk of post-treatment infections. Consequently, LLLT is becoming an integral part of endodontic practices to ensure better patient outcomes and faster recovery.[5]

## LLLT in orthodontics

This modern therapy provides notable benefits in orthodontics. It significantly reduces pain and discomfort from braces and other orthodontic devices, improving patient comfort. LLLT accelerates tooth movement by stimulating bone remodeling, potentially shortening the duration of treatment. Additionally, it helps in reducing inflammation and encouraging tissue healing, particularly useful after adjustments or in managing oral ulcers caused by braces. As a result, LLLT is increasingly being incorporated into orthodontic care for enhanced patient satisfaction and treatment efficiency.

## LLLT in implantology

This therapy is increasingly valuable in implantology, offering several key benefits. It enhances osseointegration by promoting bone attachment around dental implants, reduces postoperative pain, minimizes inflammation and edema, and accelerates tissue repair and wound healing through stimulated cellular activity and improved blood circulation. Additionally, LLLT has antimicrobial properties, reducing the risk of peri-implantitis and infections. Due to these advantages, LLLT is progressively being adopted to improve patient outcomes and the success of dental implants.[6]

## LLLT in oral lesions

It is effectively used in treating oral lesions, providing several benefits. It accelerates the healing of various oral ulcers and sores by stimulating cellular repair and enhancing blood flow. LLLT also significantly reduces pain and discomfort associated with oral lesions, offering immediate relief. Additionally, it minimizes inflammation and swelling, promoting faster recovery. With its antimicrobial properties, LLLT helps in reducing infection risks in the affected areas. These advantages make LLLT a valuable tool in managing and treating oral lesions, improving patient comfort and healing outcomes.[6]

## LLLT in periodontics

In periodontics, it aids in faster wound healing, reduces pain, and enhances outcomes of non-surgical periodontal therapy. It promotes bone and gingival

tissue healing, decreases surgery side effects, and can significantly lower inflammation and bacterial loads. Additionally, it has been shown to enhance cellular activities, such as fibroblast and osteoblast proliferation, which are critical for tissue regeneration. Understanding optimal parameters and safety measures is essential for maximizing therapeutic benefits, ensuring effective integration into routine periodontal treatments. Continuous advancements and research will further define its role, making it a valuable tool in modern dental practice.[7]

### LLLT in temporomandibular disorder

By reducing inflammation, decreasing pain, and promoting tissue repair, LLLT improves jaw function and mitigates discomfort. This non-invasive treatment enhances cellular activity and blood flow within the temporomandibular joint and surrounding muscles, offering a promising alternative to conventional temporomandibular disorder (TMD) interventions. Regular LLLT sessions can significantly contribute to the management and relief of TMD symptoms.[8]

### LLLT in management of hypersensitivity

LLLT effectively addresses tooth hypersensitivity by desensitizing dentin and promoting reparative processes. The laser application helps seal dentinal tubules, reducing fluid movement that triggers sensitivity. This non-invasive treatment offers immediate relief and long-term benefits, making it a valuable option for managing dental hypersensitivity.[9]

### Advantages and limitations of LLLT

This therapy promotes healing, reduces pain and inflammation, enhances tissue regeneration, and minimizes infection risk. However, it requires multiple sessions, costly equipment, and skilled operators, with limited efficacy for deep tissues. Despite these, LLLT remains a promising therapy for committed patients and expert application.[1,6,7]

### Future prospective

The future of LLLT in dentistry is promising. Advancements in laser technology and clinical protocols are likely to expand its applications, including enhanced periodontal therapies, improved implant integration, and accelerated orthodontic treatments. Continued research may reveal new benefits for tissue regeneration, pain management, and minimally invasive dental procedures, making LLLT an integral part of comprehensive dental care.

## Conclusion

LLLT offers numerous dental benefits, promoting healing and reducing pain with minimal side effects. Its efficacy depends on specific parameters, such as application site, wavelength, and dosage. While promising for various treatments, further research is needed to ensure safety and understand the underlying mechanisms and optimal therapeutic windows.

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### Conflicts of interest

There are no conflicts of interest.

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